In Search of Lost Time: A Theory of the Time Course of Visual Working Memory Retrieval

Some People...
The University of Melbourne

Author Note

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Abstract

The heavy-tailed distributions of errors found in continuous-outcome retrieval from visual working memory (VWM) imply that some stimuli are represented in memory with very low precision. We present a new theory of VWM retrieval in which the magnitude of retrieval errors reflects polar variability in drift rate in the circular diffusion model (CDM). The polar variability in drift rate is described mathematically by a wrapped Cauchy distribution, which characterizes the output of a statistically variable memory encoding and consolidation process that is modeled as a subordinated Brownian motion process. The infinite divisibility of the wrapped Cauchy in the CDM predicts heavy-tailed error distributions that scale as the inverse square root of set size, as predicted by a sample-size model of memory resources. We applied the Cauchy CDM to data from three continuous-outcome color VWM experiments and found it gave a good account of decision times and retrieval accuracy, except that the set-size effect in experiments with large displays was better described by a power law whose exponent exeeded that of the sample-size model. The power law is predicted by an attention weighted sample-size model in which attention prioritizes a subset of around four items during memory encoding, increasing their representational precision.

The introduction of the continuous-outcome decision task to the study of visual working memory (VWM) retrieval by Wilken and Ma (2004) gave researchers a new tool for understanding how the capacity limitations of VWM impose a bottleneck on stimulus identification processes. Prior to their work, VWM retrieval was most often studied using two-choice decision tasks — either change detection or some other kind of two-alternative, forced-choice (2AFC) or rating-scale task (Donkin et al., 2014, 2016; Luck & Vogel, 1997; Palmer, 1990; Rouder et al., 2008; Vogel et al., 2006) — and analyzed using signal detection theory or related methods. The use of the continuous-outcome task transformed the study of VWM because it highlighted the unusually large variability in the information retrieved from memory across trials and its dependence on set size (Bays et al., 2024; Ma et al., 2014). The principal empirical finding is that the task yields peaked, heavy-tailed distributions of errors, whose variability exceeds that expected from normally-distributed stimulus representations. Many cognitive processes are described successfully by assuming normally-distributed representations, but the error distributions from continuous-outcome tasks imply that something more is required theoretically. Among the competing proposals are: (1) the distributions arise from a combination of memory-based decisions and guessing, reflecting the item-capacity limited nature of VWM (Zhang & Luck, 2006); (2) they reflect across-trial variability in the precision with which items are encoded in VWM (Fougnie et al., 2012; van den Berg et al., 2012, 2014); (3) they reflect the statistical properties of neural population codes (Oberauer, 2023; Schurgin et al., 2020; Schneegans et al., 2020; Tomić & Bays, 2024); and (4) they reflect interference among items that normalizes retrieval probabilities via a Luce choice rule (Oberauer & Lin, 2017). A recent, more radical, proposal by Polyn and Woodman (2026) eschews the concept of cognitive capacity entirely and argues that the apparent VWM capacity limitations are a reflection of context drift in long-term memory. In this framework, active, or working, memory, is the part of memory that falls within the scope of the current context and VWM item limits are expressions of the context span. Their view is echoed in a recent opinion piece by Logan (2026) who argued that phenomena often attributed to capacity limitations may be better conceptualized as reflecting an interaction of attentional selection and normalization.

Most of the models cited above tacitly make the assumption that the distribution of retrieval errors in continuous-outcome tasks is a more or less veridical reflection of the variability of the memory representation in VWM — possibly augmented with some comparatively small, additional components of variability due to perceptual encoding (Tomić & Bays, 2024) or motor execution (van den Berg et al., 2014). A contrasting view of memory retrieval is provided by evidence accumulation models of decision making like the diffusion model (Ratcliff, 1978; Ratcliff et al., in press; Smith & Ratcliff, 2025), which assume that multiple sources of variability contribute to retrieval errors in memory tasks. These sources of variability include both across-trial variability in memory representations

and moment-to-moment, within-trial variability in the decision process that retrieves them. Ratcliff et al. and Smith and Ratcliff describe numerous studies in which diffusion models have been applied successfully to characterize memory retrieval decisions of various kinds. These include recognition memory, lexical decisions, and decisions about briefly-presented backwardly masked stimuli in perceptual tasks. The latter require stable stimulus representations in visual short-term memory in order to perform them because the stimulus is too brief to identify before its perceptual representation is erased by the mask (Ratcliff & Rouder, 2000; Smith et al., 2004; Smith & Ratcliff, 2009). Our aim in this article is to characterize VWM retrieval within a diffusion model framework and to show its implications for how we conceive of VWM capacity.

The principal reason for the widespread use of diffusion models in perception and memory studies is because they provide an account of both response time (RT) and accuracy, not just the latter. Since the pioneering work of Donders (1869), one of the primary aims of cognitive psychology has been to understand cognitive processes through the time taken to complete them, so much so that Luce (1986, p. 1) called RT "psychology's ubiquitous dependent variable." In contrast to the widespread use of RT methods in other parts of cognitive psychology, RT methods have been used comparatively infrequently in VWM studies. The studies of Donkin et al. (2013) and Sewell et al. (2016), who fitted VWM retrieval times using a linear ballistic accumulator model and a two-choice diffusion model, respectively, are exceptions. It is sometimes argued that because RT typically does not vary much with set size, it has little to contribute to our understanding of VWM. However, this putative invariance is something that stands in need of explanation. Moreover, RT does vary systematically with variables that affect the difficulty of retrieval decisions when they are manipulated along with set size and these relationships also stand in need of explanation. We give examples of these kinds of effects later. A more pragmatic reason for this neglect is that, until recently, there have been no models of continuous-outcome decisions comparable to those for two-choice decisions that could predict accuracy and RT.

Recently, two models of continuous-outcome decisions have been proposed based on diffusion processes: the circular diffusion model (CDM) of Smith (2016) and the spatially continuous diffusion model (SCDM) of Ratcliff (2018). These models can predict distributions of retrieval errors and joint and marginal distributions of errors and RTs. There have been other proposals subsequently, like the multiply anchored accumulation theory (MAAT) model of Kvam et al. (2023), which assumes ballistic accumulation, and an n-alternative generalization of Usher and McClelland's (2001) leaky competing accumulator model, by Hadian Rasanan et al. (2022), which can approximate continuous responding when n becomes large, but these models have not been evaluated against data in the same detail as have the CDM and SCDM. As argued by Smith et al. (2026), diffusion models have a particular claim on our attention because of their widespread success in accounting

for performance in a wide range of perceptual and memory decision tasks. We have chosen to use the CDM here because of its analytic tractability, which allows us to derive exact theoretical predictions for how the precision of memory representations and the associated retrieval errors vary with set size and other variables. There are no comparable analytic predictions for the SCDM; its predictions must be evaluated by simulation. Smith et al. (2026) recently compared the CDM and the SCDM on several perceptual tasks, using simulation methods in either instance, and showed that both models accounted for the marginal and joint distributions of errors and RT when they were equipped with suitable assumptions about the evidence entering the decision process. We would therefore expect the two models to provide broadly similar accounts of VWM retrieval decisions, because of their common diffusion process architectures. Fennell and Ratcliff (2023) applied the SCDM to data from several VWM experiments and showed that a model with both item-capacity and resource limits could fit their RT and accuracy data. However, the aims of their work were narrower than ours, because they were not concerned to develop a theory of VWM resource capacity. We discuss their work in relation to our own subsequently.

The Jones-Pewsey and Wrapped Cauchy Circular Diffusion Models

Our purpose in this article is to present a new theory of VWM capacity that we relate theoretically to drift rates in the CDM. The new theory is based, in part, on the results of an unpublished study of VWM by Smith et al. (2024/2026) that applied the CDM to two eye-movement and two mouse-movement VWM experiments in which they varied the range of memory set sizes and the difficulty of the decision task. One of the main results from the study was that the latent memory distributions, inferred from fits of the CDM to the RT and accuracy data, had an invariant shape across experimental conditions, while their dispersion varied with set size. We interpret the shape invariance as evidence of a VWM encoding and consolidation process of a particular kind, which predicts latent memory distributions of precisely this shape. Memory representations in our theory are the outcome of interacting attentional selection and VWM encoding and consolidation processes, which we model mathematically as a *subordinated Brownian motion process*. We show how the properties of the VWM memory process predict latent memory distributions with this shape and provide support for an attention-weighted sample-size model VWM capacity. We confirm the predictions of the theory in fits to experimental data.

The organization of the remainder of this article is as follows. First we describe the CDM and the extension proposed by Smith et al. (2020), the JP-CDM, in which the polar distributions of drift rate angles were modeled using a three-parameter Jones-Pewsey (JP) distribution (Jones & Pewsey, 2005; Wernicke et al., 2026). Smith et al. (2020) used the JP-CDM to model performance in a hue identification task with chromatic noise stimuli, which yields heavy-tailed error distributions, like those found in VWM tasks. The JP-CDM predicted the error distributions from this task about as well as did a two-

component encoding-failure model, in which perceptual encoding fails on some trials and responding is random. The encoding-failure process is analogous to the guessing process in two-component memory retrieval models like that of Zhang and Luck (2008) and others. Smith et al. (2024/2026) showed that their RT distribution and accuracy data were well described by a JP model in which memory precision scaled according to a sample-size model or its power law generalization. Smith and colleagues previously found support for the sample-size model in a variety of VWM tasks using two-choice decisions (Lilburn et al., 2019; Lilburn & Smith, 2020; Sewell et al., 2014, 2016). The model predicts that memory precision scales as the inverse square-root of set size, m, or, equivalently, as a power law, $m^{-\beta}$, with exponent $\beta = 0.5$. The set-size effect increases with attentionally demanding stimuli and is better described by a power law with an exponent $\beta > 0.5$. Larger exponents are predicted by an attention-weighted sample-size (AWSS) model, in which attention is allocated unequally to items during encoding (Smith et al. 2016, 2018). We report analogous results for continuous-outcome decision tasks in this article.

The JP distribution used by Smith et al. (2024/2026) independently parameterizes the location, dispersion, and shape of the latent memory distributions. They chose to use the JP distribution to characterize the latent memory representations because of its flexibility. As the shape parameter, ψ, decreases negatively from $\psi = 0$, the JP distribution changes from a bell-shaped, von Mises distribution to a highly leptokurtic distribution with extended tails. The point of departure for our theory here is the observation that, on average, our individual subject data are well-described by a JP distribution with $\psi = -1$, as detailed below. This special case of the JP distribution is a *wrapped Cauchy distribution,* which has several useful theoretical properties. Mathematically, the wrapped Cauchy is an *infinitely divisible* distribution, like the normal distribution. This means that a sum of wrapped Cauchy random variables is also distributed as a wrapped Cauchy, as is the case for the normal distribution. The infinite divisibility of the distribution allows us to derive a sample-size relationship, in which the precision of the memory representation scales inversely with the square-root of the set size, just as it does in the normal signal detection model of VWM (Palmer et al., 1990; Sewell et al., 2014).

Theoretically, the wrapped Cauchy describes the distribution of a Cauchy stochastic process, which is one member of the general class of Lévy processes (Applebaum, 2004; Sato, 2013). Mathematically, the Cauchy process arises as a result of a doubly-stochastic, subordinated Brownian motion process in which a Brownian motion diffusion process runs for a random time that is determined by the first-passage time of a second, independent, Brownian motion process. It is the doubly-stochastic nature of the subordinated Brownian motion process that allows the Cauchy CDM to predict heavy tailed memory distributions that agree with those found in fits of the CDM to experimental data. We identify the subordinated Brownian motion process with a memory encoding and consolidation process

that runs for a random time after the stimulus display has terminated, during which the variability of the memory representation continues to increase. We show here how an AWSS version of the Cauchy CDM provides a good fit to experimental data.

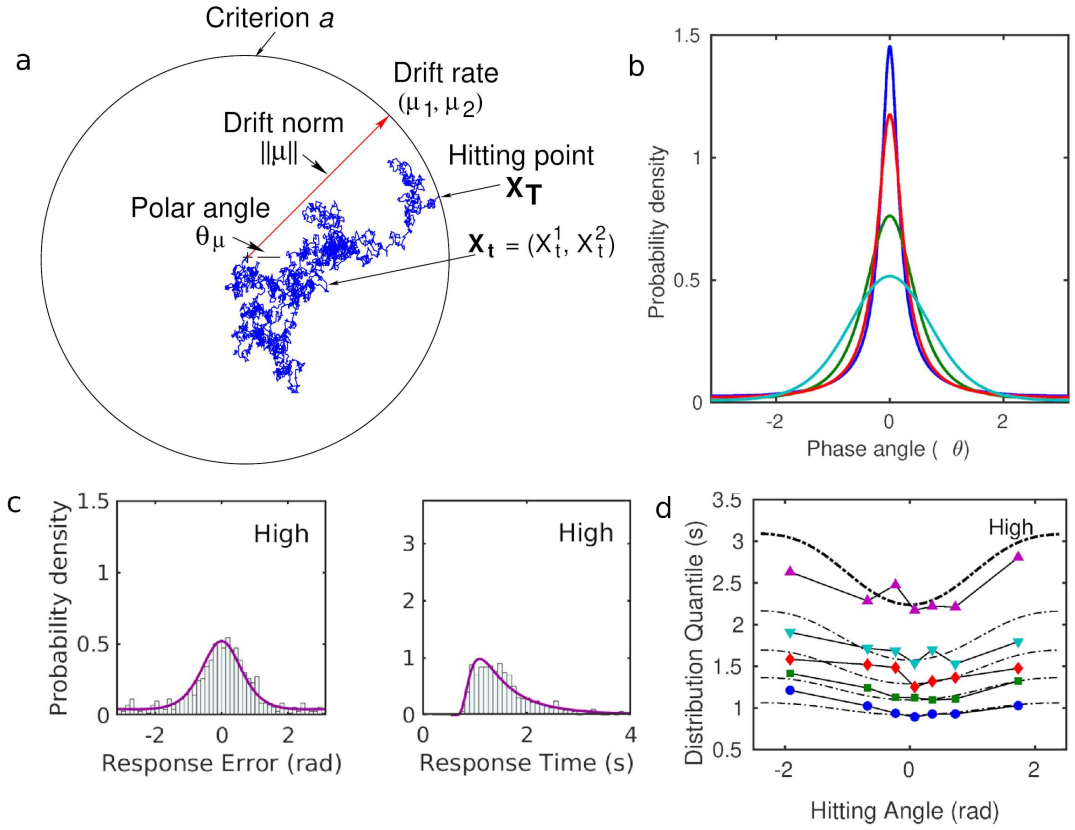


Figure 1: *Circular diffusion model.* (a) Evidence is accumulated on the interior of a disk of radius a by a 2D Wiener process, $\boldsymbol{X_t}$, with components $(X_t^1, X_t^2)'$. The point at which the process hits the bounding circle, $\boldsymbol{X_T}$, is the decision outcome and the time, T, taken to hit it is the decision time. The drift rate is vector-valued, $\boldsymbol{\mu}$, with components $(\mu_1, \mu_2)'$, with norm $\|\boldsymbol{\mu}\|$ and direction, or polar angle, θ_μ. The polar angle represents the encoded stimulus identity and the norm represents the quality of the encoded representation. (b) Jones-Pewsey distribution of variability in drift rate polar angles. The curves in the figure, from least to most peaked, are for $\psi = 0$, -0.5, -1.0, -1.25, all with $\kappa_\mu = 2$. The $\psi = 0$ case is a von Mises distribution; the $\psi = -1$ case is a wrapped Cauchy distribution. Fit of the JP-CDM to data for one subject from the high noise condition ("High") of the chromatic noise decision task of Smith et al. (2020). The data for this subject were fitted with a model with $\psi = -0.814$. (c) Marginal distributions of error and RT. (d) $Q \times Q$ plot of joint distribution showing the continuous-outcome slow-error pattern.

Figures 1a and 1b show the CDM and examples of the Jones-Pewsey distribution used to model the polar component of drift-rate variability in the JP-CDM. Figures 1c and 1d

show an example of the fit of the model to the marginal and joint distribution of errors and RT to data for one subject from the high noise condition of the chromatic noise color task of Smith et al. (2020a). The marginal distribution of errors shows the heavy tails that Smith et al. attributed to an encoding failure process. The joint distribution of errors and RT is shown as a $Q \times Q$ plot (Smith et al., 2020, 2022, 2024/2026), in which selected quantiles of the RT distribution are plotted on the y-axis against quantiles of the error distribution on the x-axis. The $Q \times Q$ plot shows a characteristic dished profile, in which the least accurate responses are also the slowest, which is the continuous-outcome counterpart of the slow-error pattern found in two-choice decisions.

In the CDM, evidence is accumulated by a 2D Wiener, or Brownian motion, diffusion process, $\boldsymbol{X_t} = (X_t^1, X_t^2)'$, on the interior of a disk, whose bounding circle, of radius a, represents the decision criterion for the task. As in the two-choice diffusion model, the decision criterion represents the amount of evidence needed to make a decision and is assumed to be under the decision-maker's control. The superscripts on $\boldsymbol{X_t}$ index the horizontal and vertical components of the process and the prime denotes matrix transposition. The evidence accumulation process is described by a vector-valued stochastic differential equation,

$$\boldsymbol{dX_t} = \boldsymbol{\mu}\, dt + \boldsymbol{\sigma}\, \boldsymbol{dW_t}, \tag{1}$$

which describes the change in the process during a small interval of length dt.

The irregular trajectory, or sample path, in Figure 1a represents one realization of the process $\boldsymbol{X_t}$ on a single experimental trial. In our notation, bold-faced symbols denote vector-valued functions and processes, upper-case Roman letters denote random variables, and lower-case Greek letters denote constants. Equation 1 states that $\boldsymbol{dX_t}$ is the sum of a deterministic part, $\boldsymbol{\mu}\, dt$, and a random part, $\boldsymbol{\sigma}\, \boldsymbol{dW_t}$, each consisting of two components. The deterministic part is a vector-valued drift rate, $\boldsymbol{\mu} = (\mu_1, \mu_2)'$, shown in the figure as an arrow at $\pi/4$ rad. (45°), which represents the information in the stimulus. The random part, $\boldsymbol{dW_t} = (dW_t^1, dW_t^2)'$, is the differential of a 2D Wiener, or Brownian motion, process. In polar coordinates, the drift rate has length or norm, $\|\boldsymbol{\mu}\| = \sqrt{\mu_1^2 + \mu_2^2}$ and polar angle, or direction, $\theta_\mu = \text{arctan2}(\mu_2, \mu_1)$, where arctan2 is the four-quadrant arctangent function. The drift rate norm and polar angle represent, respectively, the average rate at which evidence accumulates towards the boundary and the direction in which it does so. The rate at which the process diffuses towards the boundary is represented by a dispersion matrix, $\boldsymbol{\sigma} = \sigma \boldsymbol{I}$, where $\boldsymbol{I}$ is a 2×2 identity matrix and σ is the infinitesimal standard deviation. Its square, σ^2, is the diffusion coefficient.

At the beginning of a trial, the process starts at the center of the disk, $\boldsymbol{X_0} = (0, 0)'$, and on presentation of a stimulus, diffuses until it hits a point on the bounding circle. The hitting point is denoted by $\boldsymbol{X_T}$, where our use of capitalized symbols for both the process and the time subscript indicates that the hitting point and hitting time are both random

variables. The components of the hitting point are $X_T^1 = a\cos\theta$ and $X_T^2 = a\sin\theta$, where θ is its polar angle. The most likely place for the process to hit the boundary is the point that corresponds to the polar angle of the drift rate vector. An observer who reports the hitting point as the decision about the stimulus identity is a maximum likelihood observer (Smith, 2016; Smith et al., 2020).

To characterize the joint distribution of decision outcomes and decision times, we denote by $d\tilde{P}_t(\theta)$ the joint probability density that a 2D Wiener diffusion process with drift rate $\boldsymbol{\mu}$ hits the bounding circle a at the point $\boldsymbol{X_T} = (a\cos\theta, a\sin\theta)'$ at a random time T. The density is the product of two components, $\boldsymbol{Z_t}(\boldsymbol{X_T})$ and $dP_t(a)$,

$$d\tilde{P}_t(\theta) = \boldsymbol{Z_t}(\boldsymbol{X_T})dP_t(a). \tag{2}$$

The component $dP_t(a)$ is the probability density that a 2D, zero-drift process, started at the center of the circle at time zero, hits a point on its circumference at time t. It has the form

$$dP_t(a) = \frac{\sigma^2}{2\pi a^3}\sum_{k=1}^{\infty}\frac{j_{0,k}}{J_1(j_{0,k})}\exp\left(-\frac{j_{0,k}^2\sigma^2}{2a^2}t\right). \tag{3}$$

Equation 3 is obtained from the first-passage time probability density function for the 1D Euclidean distance, or Bessel, process, which characterizes the Euclidean distance of a zero-drift Wiener process from its starting point (Hamana & Matsumoto, 2013). If a zero-drift Wiener process starts at the origin then its first-passage time distribution through a circular boundary will be circularly symmetrical and identical, up to a scaling factor of $2\pi a$, to that of the 1D Bessel process.

The function $J_1(x)$ in Equation 3 is a first-order Bessel function of the first kind (Abramowitz & Stegun, 1965, p. 360) and the $j_{0,k}$ terms are the zeros of a Bessel function of the first kind of order zero, $J_0(x)$, that is, they are the points at which the function crosses the x-axis. The terms $J_1(j_{0,k})$ in the denominator represent values of $J_1(x)$ evaluated at the zeros of $J_0(x)$. The term $dP_t(a)$ is a function of time only and is independent of the hitting point on the criterion circle. Mathematically, it depends only on the diffusion coefficient, σ^2, and the radius of the criterion circle, a. The distribution of hitting points on the criterion circle is characterized by the second term, $\boldsymbol{Z_t}(\boldsymbol{X_T})$, given by the Girsanov change-of-measure theorem (Karatzas & Shreve, 1991), which takes the form

$$\boldsymbol{Z_t}(\boldsymbol{X_T}) = \exp\left[\frac{1}{\sigma^2}(\boldsymbol{\mu}\cdot\boldsymbol{X_T}) - \frac{1}{2\sigma^2}\|\boldsymbol{\mu}\|^2 t\right], \tag{4}$$

where $(\boldsymbol{\mu}\cdot\boldsymbol{X_T}) = a\mu_1\cos\theta + a\mu_2\sin\theta$ is the dot product of the drift rate vector and the vector of coordinates of hitting points on the boundary. The function $\boldsymbol{Z_t}(\boldsymbol{X_T})$ factorizes into a product of two exponential terms: one that depends on time and is independent of the hitting point and another that depends on the hitting point and is independent of time.

A striking property of the CDM is it predicts that decision outcomes will be von Mises distributed. This property is important theoretically because many models in the VWM literature represent error distributions as mixtures or weighted sums of von Mises components (Bays et al., 2009; Oberauer & Lin, 2017; van den Berg et al., 2012, 2014; Zhang & Luck, 2008). The individual von Mises components in such models are typically assumed rather than derived from more basic principles. In contrast, the von Mises distribution in the CDM arises as the expression of evidence accumulation by a diffusion process. The von Mises distribution has a density function $f(\theta; \kappa, \varphi)$, where κ is a precision or concentration parameter and φ is the distribution center,

$$f(\theta; \kappa, \varphi) = \frac{e^{\kappa \cos(\theta - \varphi)}}{2\pi I_0(\kappa)}, \tag{5}$$

and $I_0(\kappa)$ is a modified Bessel function of the first kind of order zero. The predicted error distribution in the CDM is obtained by writing the coordinates of the drift rate vector as $\mu_1 = \|\boldsymbol{\mu}\| \cos\varphi$ and $\mu_2 = \|\boldsymbol{\mu}\| \sin\varphi$ and then evaluating the dot product on the right-hand side of $\boldsymbol{Z_t}(\boldsymbol{X_T})$ in Equation 4 as

$$\exp\left[\frac{1}{\sigma^2}\left(\boldsymbol{\mu} \cdot \boldsymbol{X_T}\right)\right] = \exp\left[\frac{a\|\boldsymbol{\mu}\|}{\sigma^2}\left(\cos\theta\cos\varphi + \sin\theta\sin\varphi\right)\right]. \tag{6}$$

The term in parentheses on the right-hand side of Equation 6 is $\cos(\theta - \varphi)$ and the right-hand side of the equation as a whole is the numerator of a von Mises distribution with precision

$$\kappa = \frac{a\|\boldsymbol{\mu}\|}{\sigma^2}. \tag{7}$$

The modified Bessel function in the denominator of Equation 5, multiplied by 2π, makes the function a proper probability distribution. It is obtained by integrating the remaining terms in Equation 4 over outcomes and time, as described by Smith (2016). Equation 7 provides a theoretical decomposition of the von Mises precision into cognitive components of the decision process. Predicted precision is equal to the decision criterion multiplied by the drift rate norm divided by the diffusion coefficient; that is, it is equal to the amount of evidence used to make a decision multiplied by the quality of evidence in the stimulus divided by the noisiness of the decision process.

When there is across-trial variability in drift rates, Equation 4 is replaced by an expression $\bar{\boldsymbol{Z}}_{\boldsymbol{t}}(\boldsymbol{X_T})$, whose form depends on the how drift rates are assumed to vary. The overbar notation suggests marginalization or averaging and implies that $\bar{\boldsymbol{Z}}_{\boldsymbol{t}}(\boldsymbol{X_T})$ is obtained by integrating $\boldsymbol{Z_t}(\boldsymbol{X_T})$ over a suitably chosen distribution of drift rates. Smith and Corbett (2019) derived an expression for $\bar{\boldsymbol{Z}}_{\boldsymbol{t}}(\boldsymbol{X_T})$ for a 4D (hyperspherical) diffusion process when the drift rate, $\boldsymbol{\mu}$, has a multivariate normal distribution with independent components. The 2D bivariate case is given in Equation 9 of Smith et al. (2020). Smith (2019) generalized

the latter result to a model in which $\boldsymbol{\mu}$ has a bivariate normal distribution with correlated components, which he related theoretically to the general recognition theory of Ashby and Townsend (1986). However, neither independent nor correlated drift-rate variability allow the model to predict heavy-tailed error distributions like the one in Figure 1c.

As an alternative, Smith et al. (2020) proposed that the polar, or angular, component of drift rate follows a JP distribution. Unlike the normal distribution, for some values of its shape parameter, the JP distribution has heavy tails, which allowed Smith et al. to successfully model the error distributions on encoding-failure trials in their data. If the CDM is in canonical orientation, so that the mean drift rate vector points along the positive x-axis, then the first component of drift rate, μ_1, represents the *radial* component of drift rate and the second component, μ_2, represents the *tangential*, or polar, component. Radial variability represents across-trial variability in the encoded stimulus quality (the drift rate norm); tangential variability represents variability in the encoded stimulus identity (the drift rate polar angle). The JP distribution has a probability density function

$$f(\theta;\kappa_\mu,\varphi,\psi)=\frac{[(\cosh(\kappa_\mu\psi)+\sinh(\kappa_\mu\psi)\cos(\theta-\varphi)]^{1/\psi}}{2\pi P_{1/\psi}(\cosh(\kappa_\mu\psi))}. \tag{8}$$

Examples of this distribution are shown in Figure 1b. In Equation 8, θ is the polar angle of the drift rate vector, φ is its mean, κ_μ is the precision, and ψ is a parameter that controls the distribution shape. The term $P_{1/\psi}(\cdot)$ in the denominator, which makes the function a probability distribution, is an associated Legendre function of the first kind of degree $1/\psi$ and order 0 (Gradshteyn & Ryzhik, 2007, 8.7–8.8).

Smith et al. (2024/2026) used the JP distribution to characterize the latent memory distribution in fits of the CDM to colour VWM experiments in which they varied the difficulty of the memory task and the decision task. Their experiments used small-N psychophysical designs (Smith & Little, 2018) in which they collected several thousand experimental trials from each of a small number of subjects. These designs provided the high levels of measurement precision needed to characterize joint distributions of errors and RT in fits of the CDM, like the one in Figure 1d. Two of the experiments used a small range of set sizes, $m = 1, 2, 3, 4$ and two used a large range, $m = 1, 2, 4, 6$. A small range of set sizes was used to minimize the effects of any item-capacity limits on retrieval performance. The principal aim of the study was to identify the scaling law that characterizes the resource capacity limits of VWM, which is easier to do when the influence of guessing is limited. Two of the experiments also varied the difficulty of the decision task by varying the color saturation of the stimuli and the color wheel on which responses were made. Smith et al. found their data were well-described by a JP-CDM in which the shape parameter, ψ, remained invariant across experimental conditions (which varied in number from four to 12, depending on the design) and in which the latent memory precision in the

CDM, controlled by the precision parameter, κ_μ, followed a sample-size model or power law in set size.

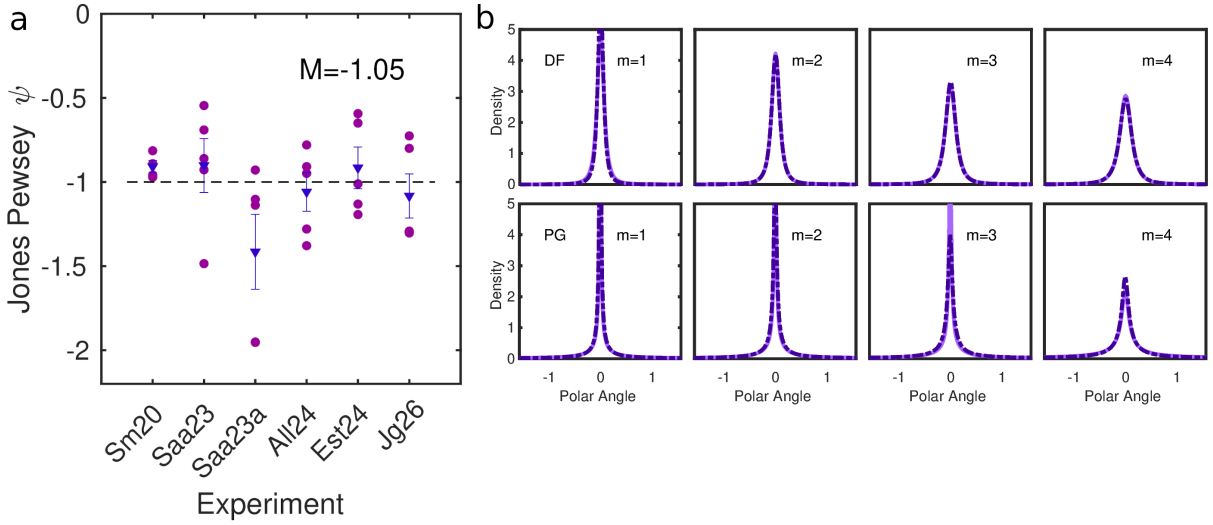


Figure 2: *Estimated Exponents and Example Drift Rate Distributions for the JP-CDM.* (a) The points are estimated exponents, ψ, of the JP distribution of drift rates in the JP-CDM from 29 individual subjects in six experiments designated by the identifying codes on the x-axis. The error bars are ± 1 SEM. The mean across all estimates is $M = -1.05$. A JP distribution with an exponent of $\psi = -1$ is a wrapped Cauchy distribution. (b) Estimated JP drift rate distributions for two subjects from Experiment 1 of Smith et al. (2024/2026, Saa23). The solid lines are for a free-κ_μ model; the broken lines are for a sample-size κ_μ model. The estimated JP exponents for DF were -0.69 and -0.68 for the free and constrained models, respectively; the exponents for PG were -1.48 and -1.69.

Smith et al. (2024/2026) allowed the JP shape parameter, ψ, to vary freely for individual subjects and showed the resulting JP-CDM model provided a good account of the marginal and joint distributions of errors and RT. The estimated drift rate distributions are plotted in their Figures 7 and 12. Smith et al (2020, Figure 8B) plot the drift rate distributions for the chromatic noise task. Figure 2b shows the polar components of the drift rate distributions for two subjects from Saa23 with different values of ψ. However, there is an important regularity in the estimates, not remarked on by them, which is brought out in Figure 2a. The figure shows ψ estimates from 29 individual subjects in six experiments in which the data were fitted with the JP-CDM. The experiments are designated using the identifying codes on the x-axis. Sm20 is the chromatic noise decision task of Smith et al. (2020). Saa23 and Saa23a were eye-movement VWM tasks using saturated color stimuli and a saturated color wheel, with a small range, $m_{\max}= 4$, and large range $m_{\max}= 6$ of set sizes, respectively. All24 and Est24 were mouse-movement VWM tasks, using a large

range and a small range of set sizes, $m_{\max}$= 6 and $m_{\max}$= 4, respectively, which varied the difficulty of the decision task by varying the color saturation of the stimuli and the color wheel. Jg26 is a replication of All24 that used a blocked rather than a randomized manipulation of color saturation. The shape parameters of the latent memory distributions in the JP-CDM are difficult to estimate reliably, independently of their dispersions, because the distributions are inferred rather than observed. They determine the properties of the observed error distributions only indirectly, via the distribution of drift rates in the CDM, which itself contributes variability to the error distribution. The estimation problem is particularly challenging for values in the range of theoretical interest for continuous-outcome decisions, in which the best-fitting drift-rate distributions are sharply leptokurtic. There is appreciable scatter in the individual ψ estimates in Figure 2a, most notably for the eye-movement experiment Saa23a, in which precision was lower than in the other experiments — likely because of measurement noise in the eye-movement recordings. Despite these caveats, on average, the estimates tend to cluster around a mean value of -1. The grand mean, across all of the estimates in the figure, is $M = -1.05$.

The value of $\psi = -1$ is a theoretically significant one because it identifies one of the distinguished, special cases of the JP distribution, namely, the wrapped Cauchy distribution (Jones & Pewsey, 2005, Case 3). The theory we develop in this article is based on the properties of this distribution. Like other wrapped distributions, such as the wrapped normal and t distributions, the wrapped Cauchy distribution is obtained by taking the Cauchy distribution on a line and wrapping it around a circle (Mardia & Jupp, 2000; Wernicke et al., 2026). Wrapped distributions are not particularly convenient to deal with theoretically or empirically because they lead to representations in the form of infinite series, which must be truncated in computations. This is why the von Mises distribution is often preferred to the wrapped normal, which it closely approximates, in applications. Surprisingly, however, the wrapped Cauchy also has a simple closed-form representation, which makes it refreshingly easy to work with, both theoretically and in fitting models to data.

The Cauchy distribution on the real line has a notorious property that is often used as a source of counterexamples in statistics, namely, its variance is not finite. Consequently, there is no central limit theorem for sums of Cauchy random variables. An estimate of a population parameter based on an average of Cauchy random variables will be no more acccurate than one based on any of the variables individually. This property might seem to make the Cauchy an unpromising starting point for a theory of VWM capacity, but, as we will see below, despite its unpromising origins, the wrapped Cauchy has nice theoretical properties that make it attractive.

Apart from its infinite series representation (Mardia & Jupp, 2000, p. 51), the probability density function of the wrapped Cauchy distribution can also be expressed in the

simple closed form as

$$f(\theta; \kappa_\mu, \varphi) = \frac{\sinh \kappa_\mu}{2\pi[\cosh \kappa_\mu - \cos(\theta - \varphi)]}. \tag{9}$$

We use the same symbols to parameterize the wrapped Cauchy distribution as we used for the JP distribution in Equation 8, but κ_μ now functions as a dispersion rather than a concentration parameter (i.e., the distribution becomes more dispersed rather than more concentrated as κ_μ gets larger). Jones and Pewsey (2005) give an alternative parameterization of the wrapped Cauchy in terms of $\tanh(\kappa_\mu/2)$ in which κ_μ functions as a concentration parameter, as it does in Equation 8. For the purpose of implementing the AWSS model using a wrapped Cauchy distribution in the CDM, we found it more convenient to treat κ_μ as a dispersion parameter because of its simple relationship to the circular variance and standard deviation, described below.

The statistics of distributions on circular spaces do not behave like those of distributions on a real line and, unlike the Cauchy distribution on the real line, the wrapped Cauchy distribution has finite variance. For a collection of n randomly-oriented unit vectors with polar angles R_i, the circular variance and circular standard deviation (CSD) are functions of $\bar{R}$, the resultant vector that is obtained by decomposing the vectors into their sine and cosine components, averaging the components, $(\sum_i \cos R_i)/n$ and $(\sum_i \sin R_i)/n$, and then computing the length of the average by Pythagoras's theorem (Mardia & Jupp, 2000; Wernicke et al., 2026). The circular variance and standard deviation are defined as $1-\bar{R}$ and $(-\log \bar{R}^2)^{1/2}$, respectively, where "log" is the natural logarithm. The intuition behind these measures is that the length of the averaged vectors gets closer to zero as their distribution becomes more uniform, which is reflected in value of $\bar{R}$. Bays and colleagues have used the reciprocal of the CSD as a measure of recall precision in continuous-outcome VWM tasks (Bays et al., 2011). The theoretical CSD and variance are defined analogously to the empirical statistics for a sample, with expectations replacing averaging. For distributions like the JP and wrapped Cauchy distributions, which are symmetrical around $\theta = \varphi$ (zero in our applications), the resultant reduces to $E_\theta[\cos(\theta - \varphi)]$, where the expectation is taken with respect to $f(\theta)$, the density function of the random variable θ, The other component of the resultant, $E_\theta[\sin(\theta - \varphi)] = 0$ because the product $f(\theta - \varphi)\sin(\theta - \varphi)$ is an odd function and integrates to zero. For the general JP distribution, the theoretical variance and CSD are expressed as ratios of associated Legendre functions, which are not particularly convenient to work with, but for the wrapped Cauchy they have a strikingly simple form. The circular variance and CSD are

$$v_R = 1 - e^{-\kappa_\mu} \tag{10}$$

and

$$CSD = \sqrt{2\kappa_\mu}, \tag{11}$$

respectively. The simplicity of the representations in Equations 10 and 11 makes them convenient for theoretical work, particularly for characterizing the scaling laws that characterize VWM capacity, which are central to the theory we develop here. In the wrapped Cauchy CDM (hereafter the WC-CDM) the wrapped Cauchy distribution (equivalently, a JP distribution with $\psi = -1$) is used to model the polar angle of drift rate variability in the CDM).

VWM Consolidation as Subordinated Brownian Motion

As in the JP-CDM, drift rate in the WC-CDM has two components: a radial component, which describes across-trial variability in the drift rate norm, which is assumed to be normally distributed, and a polar or angular component, which is assumed to be a wrapped Cauchy distribution. The statistical properties of the wrapped Cauchy distribution allow the CDM to predict the heavy-tailed error distributions found in VWM. The estimates of ψ in Figure 2 provide an empirical justification for using the WC-CDM as a model of VWM retrieval, which we might prefer to the JP-CDM purely on grounds of parsimony. However, our aim is to provide a deeper, theoretically motivated characterization of why a Cauchy process should arise arise psychologically. Our analysis is based, first, on identifying the kind of physical processes that gives rise to Cauchy processes mathematically and, second, on identifying the kind of psychological processes that might exhibit these particular mathematical properties. Mathematically, the wrapped Cauchy distribution arises as the result of a particular kind of doubly-stochastic process called a subordinated Brownian motion process, which we identify here with the properties of a time-varying VWM encoding and consolidation process. The properties of the resulting model are shown in Figure 3.

The terms "encoding" and "consolidation" do not have settled meanings in the literature and are used in different ways by different authors. For our purposes, it is convenient to distinguish between processes that occur while the stimulus is physically present, which we term "encoding," and processes that may continue for a period after the stimulus has terminated, which we term "consolidation." In general, the quality of the encoded stimulus representation will increase with exposure duration (Sewell et al., 2014; Smith et al., 2018; Vogel et al., 2006) up to some asymptotic value, which Bays et al. (2011), estimated to be around 300 ms in VWM for orientation. Consolidation may continue for an extended period after this point and, according to literature reviewed by Ricker and Hardman (2017), may take anywhere between 400 ms to 1600 ms. They argue that the function of consolidation is to stabilize fragile memory traces and present evidence that consolidation effects found experimentally are due to a failure to do so for some items. Much of the existing research on consolidation, summarized in their article, focuses on its attention demanding, limited-capacity aspects, and its susceptibility to interference. We view consolidation in the same way as do Ricker and Hardman, as a process whose end point is the formation of a stable VWM trace, but our theory makes two specific assumptions about this process.

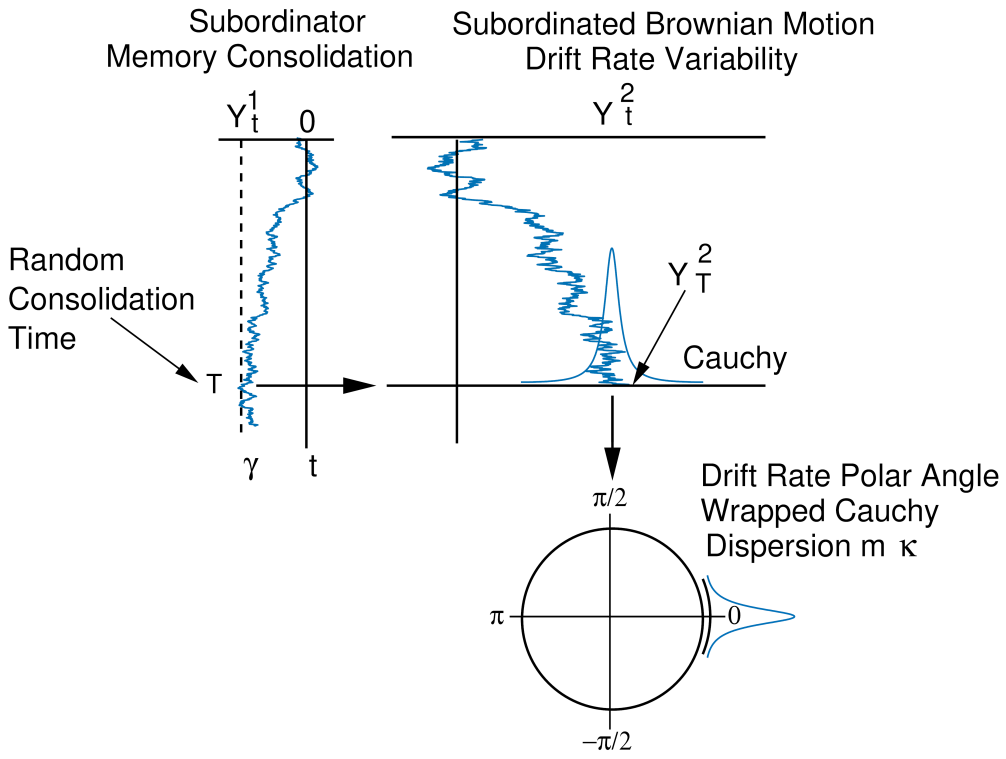


Figure 3: *Wrapped Cauchy Distribution of Drift Rate Variability.* The Cauchy distribution on the line is obtained from a subordinated Brownian motion process that models the time course of VWM encoding and consolidation. The first process, the subordinator, Y_t^1, is a zero-drift, unit-variance Brownian motion that runs until a random time, T, when it hits a boundary γ. The hitting time is the time at which the encoded representation in memory becomes fixed. The polar angle of drift rate variability is described by a second, nonzero drift, Brownian motion process, Y_t^2. The value of Y_T^2, at the time when the first process hits the boundary is Cauchy distributed on the line. The distribution of drift rate polar angles is a wrapped Cauchy distribution, obtained by wrapping the Cauchy distribution around a unit circle. The dispersion of the wrapped Cauchy scales as $m\kappa$, where m is the number of items in the display and κ is the dispersion for a single item.

First, consolidation times are random and variable and, for some items, may be significantly extended. Second, the longer consolidation is delayed, the poorer — in the sense of the more variable — will be the final VWM representation. In the Cauchy CDM, the effects of this variability are characterized by the polar angle of the drift rate distribution.

The idea that memory consolidation times are variable is one with a long history in the literature. Often, consolidation times are assumed to be exponentially distributed. In the Theory of Visual Attention (TVA) of Bundesen and colleagues, for example, items are "pigeonholed" — in Broadbent's (1971) sense of the term — after an exponentially

distributed delay (Shibuya & Bundesen, 1988; Bundensen, 1990; Bundesen et al., 2005, 2011). In the early form of TVA (Bundesen, 1990), pigeonholing was equated with identification of the stimulus, but in the later form (Bundesen et al., 2011), it is identified with the input to an evidence accumulation decision process, which aligns more closely with our theory here. Models with exponentially distributed processing times have a long history in cognitive psychology, in part because of their analytic tractability (Fisher, 1982; Liu, 1996; Townsend, 1972, 1976), but, for our purposes, the exponential distribution has the wrong properties. Instead, we assume that the time course of consolidation is governed by the first-passage time of a zero-drift, unit variance, Brownian motion diffusion process. This process, denoted Y_t^1 in Figure 3, determines the time required for an item to be consolidated in VWM. In the stochastic processes literature this process is called a *Lévy subordinator* (Applebaum, 2004; Sato, 2013). Its first passage time density is

$$g_T(t) = \frac{\gamma}{\sqrt{2\pi t^3}} \exp\left(-\frac{\gamma^2}{2t}\right), \tag{12}$$

where the random variable T is the consolidation time and the decision boundary, γ, is a parameter of the model (Cox & Miller, 1965, p. 221). The zero-drift, first-passage time process has the property that it terminates at the boundary with probability one, but its expected value is not finite (Karatzas & Shreve, 1991, p. 96). Psychologically, this means that the consolidation process for an item in VWM is guaranteed to terminate eventually, but not necessarily within the retention interval of an experiment. This means that the VWM representations for some items may not have stabilized by the time they are probed for report.

The subordinator controls a second, independent Brownian motion process, Y_t^2, which encodes the information in the stimulus, and which runs until the random time T. Its value at time T determines the polar angle of the memory representation. Because Y_t^2 is a Brownian motion process, its variance grows linearly with time. Consequently, the longer consolidation is delayed, the more variable the representation becomes. It is straightforward to show, as we do in Appendix A, that the distribution of Y_t^2, when it is stopped at a random time, T, is Cauchy. In tasks that use circular stimulus spaces, like those we consider here, the Cauchy distribution on the line is wrapped around the circle to yield the wrapped Cauchy distribution, as shown in Figure 3. The doubly-stochastic nature of the subordinated Brownian motion process is the key to obtaining the wrapped Cauchy distribution of VWM in the CDM.

The Sample-Size Model in Two-Choice and Continuous-Outcome Decisions

The sample-size model is a statistical theory of cognitive resources that Lilburn and Smith (2020) traced back to the origins of signal detection theory (Swets et al., 1959). It assumes that the resources available to represent stimuli comprise a set of noisy evidence samples that are recruited during stimulus encoding and which are distributed among the

items in the display to represent them. Its signature is an inverse square-root law in set size: In two-choice tasks, in which performance is measured using signal detection theory d', it predicts that

$$d'_m = \frac{d'_1}{\sqrt{m}}, \tag{13}$$

for an m-item display, where d'_1 is sensitivity for a display with a single item. The model has found wide applicability in intramodal and crossmodal divided attention studies (Bonnel & Hafter, 1998; Bonnel & Miller, 1994; Lindsay et al., 1967; Taylor et al., 1967), divided attention (Eckstein et al., 2000; Palmer, 1990; Shaw, 1980, 1982, 1984), and object-based attention (Corbett & Smith, 2017) and, more recently, VWM (Palmer, 1990). Sewell et al. (2014) found it gave a good account of VWM capacity for oriented grating patches and Sewell et al. (2016) showed that a diffusion model in which the drift rates followed the sample-size model gave a good account of both accuracy and RT distributions. The slots-plus-averaging model of Zhang and Luck (2008) is a form of coarse-quantized sample-size model, in which the number of samples is equal to the number of slots in memory. However, the coarse-quantized model performs more poorly than competitor models (Fougnie et al., 2012; Oberauer & Lin, 2017; van den Berg et al., 2014) and has the unattractive property that it requires a complicated bookkeeping scheme to implement its proposed storage and retrieval process (Sewell et al., 2014; Smith et al., 2024/2026).

The inverse square-root law in Equation 13 relies theoretically on the infinite divisibility of the normal distribution. Infinite divisibility means that the average of a set of normal random variables is itself normally distributed with a standard deviation equal to the standard error of the mean. The sample-size model assumes a finite number of evidence samples are available at the end of the stimulus encoding period to represent items in memory, which are distributed equally among items. If there are n evidence samples distributed among m items, such that each item is allocated n/m samples to represent it, then the precision of the representation decreases inversely as $\sqrt{m}$, yielding Equation 13.

A problem faced by Smith et al. (2024/2026) in attempting to extend the sample-size model to continuous-outcome decisions is that the JP distribution is not infinitely divisible, so there is no natural inverse square-root law associated with it comparable to that in Equation 13. Instead, Smith et al assumed a sample-size relationship on its CSD and used it to characterize the set-size effect in their data. Although this approach provided a characterization of the empirical set-size effect paralleling the one in two-choice tasks, it lacks the formal justification of the sample-size relationship in the normal signal detection model.

In contrast to the JP distribution, the Cauchy distribution *is* infinitely divisible, as is easily shown from the properties of its characteristic function (the Fourier transform of its probability density function), as we show in Appendix A. The wrapped Cauchy

distribution, which is obtained by wrapping the Cauchy distribution on the line around the circle, inherits its scaling properties. This means that a sum of m wrapped Cauchy random variables with dispersion parameter κ_μ will be distributed as a wrapped Cauchy with dispersion parameter $m\kappa_\mu$ and CSD $\sqrt{2m\kappa_\mu}$. This property forms the basis for a sample-size model of continuous-outcome decisions in the WC-CDM. The explicit sample-size model in the WC-CDM allows us to derive expressions for generalizations of the model in which evidence samples are allocated unequally across items. This assumption leads to the attention weighted sample-size (AWSS) model (Smith et al., 2016, 2018). The AWSS model predicts a power law set-size effect rather than an inverse square-root law and often better characterizes data when attentionally demanding stimuli are used.

The AWSS and the Power Law Models

The AWSS model of Smith et al. (2018) assumed that a single item in the display captures attention at random and receives a disproportionate share c_a, $0 < c_a < 1$, of the available evidence samples. The remaining evidence samples are divided equally among the other $m - 1$ items in the display. Sensitivity in an m-item display is then a mixture of sensitivity to a single attended item with weight $1/m$ and $m - 1$ unattended items with weights $(m - 1)/m$. Instead of an inverse square-root law, the model predicts a power law whose exponent depends on the extent to which c_a exceeds $1/m$, that is, the extent to which the allocation of evidence samples to items is unequal. Smith et al. showed that the magnitude of the exponent was predicted by the attention demands of the stimulus used in the VWM task, as determined by external criteria — primarily, the degree to which the stimuli engage parallel or serial processing in visual search (Thornton & Gilden, 2007). The idea that a single item captures attention and receives a disproportionate share of resources when processing attention-demanding stimuli is supported by the results of Smith et al. (2016) who found a large primacy effect and no recency effect with sequential presentation in a phase discrimination VWM task. Subjects in the phase discrimination task must decide whether Gaussian-filtered, luminous disks are darker on the left or the right side. The results of Thornton and Gilden suggest that visual search for phase targets is strongly serial. Smith et al. found that a diffusion model in which the drift rates followed the AWSS model provided a good account of the accuracy and RT distributions in this task, extending the results for the sample-size model of Sewell et al. (2016).

Here we considered a more general form of the AWSS model, in which there is some number, k_m, of attended stimuli, rather than just one. We were motivated to consider the more general model after finding the single-item version of the model failed to account for some aspects of the error distributions in continuous-outcome decisions, as we discuss subsequently. We found it convenient to parameterize the continuous-outcome AWSS model in terms of attention weights rather than capacity fractions. Because only the relative weights for attended and unattended items matter when the model is expressed in this way, we

arbitrarily set the weight for attended items to 1 and the weights for unattended items to u, which is a parameter of the model. In a display of m items there are $m_a = \min(m, k_m)$ attended items and $m_u = \max(m - m_a, 0)$ unattended items. The capacity fractions for attended and unattended items, c_a and c_u are then

$$\begin{aligned} c_a &= \frac{1}{m_a + u m_u} \\ c_u &= \frac{u}{m_a + u m_u}. \end{aligned} \tag{14}$$

The dispersion parameters of the wrapped Cauchy distribution in the CDM are $\kappa_a = \kappa/c_a$ and $\kappa_u = \kappa/c_u$, respectively, and the weighted sum $m_a c_a + m_u c_u = 1$, as required. With this scaling, an unattended item weight of $u = 0.5$ means that the circular standard deviation of the drift rate for an unattended item will be $\sqrt{2}$ greater than that for an attended item. The predicted error distributions in the CDM are mixtures of error distributions for attended and unattended stimuli with mixing proportions m_a/m and m_u/m.

In addition to the wrapped Cauchy distribution, which describes the polar angle of drift rate variability in the CDM, the radial component of drift rate also varies. Radial variability is analogous to the drift-rate variability in the two-choice model and allows the model to predict the slow-error pattern of RTs in Figure 1d. We follow Smith et al. (2020) and assume the radial component is normally distributed with standard deviation η. This allows us to use the analytic expression for $\bar{\boldsymbol{Z}}_{\boldsymbol{t}}(\boldsymbol{X}_{\boldsymbol{T}})$ from Smith and Corbett (2019), which appreciably improves the model's computational efficiency. The marginal effects of the wrapped Cauchy distribution of drift rate polar angles must be evaluated by numerical integration.

Theoretically, the JP-CDM and WC-CDM are related to the variable precision models of Fougnie et al. (2012) and van den Berg et al. (2012, 2014), in which the precision of VWM representations varies randomly across trials. Fougnie et al. modeled the variability as a wrapped t-distribution; van den Berg et al. modeled it as a gamma-distributed mixture of von Mises distributions whose dispersion parameter followed a power law. These models predict heavy-tailed error distributions because they assume the variability of memory representations exceeds that of the normal distribution. Fougnie et al. reported that their model, when augmented with a guessing process, performed better than a fixed-precision model with guessing. Van den Berg et al. reported that variable precision models performed better than both fixed precision and slots-plus-resources models, including the slots-plus-averaging model of Zhang and Luck (2008). They found that a model in which the number of items encoded into VWM across trials was Poisson distributed and the remaining items were guessed was marginally better than a model in which all of the items were encoded. The JP-CDM and WC-CDM can be viewed as variable precision models in which the latent VWM distributions rather than the empirical error distributions vary

across trials. Harlow and Donaldson (2013) found that error distributions in a continuous-outcome source memory task were well-described by a mixture model comprising a wrapped Cauchy distribution and a uniform distribution. Unlike their model, in which the wrapped Cauchy distribution provided an empirical characterization of the distribution of errors, which was essentially atheoretical in nature, the wrapped Cauchy in the CDM characterizes the latent VWM representations and arises as a consequence of VWM consolidation.

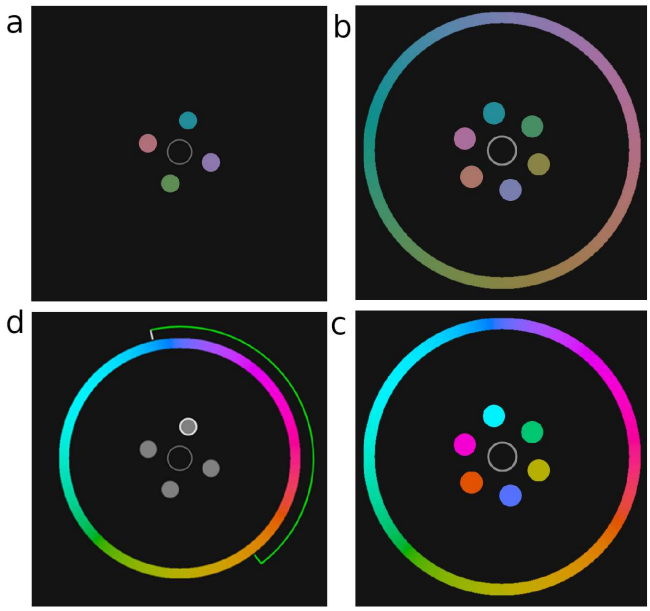


Figure 4: *Stimulus displays for Experiments 1 to 3.* All experiments used colored stimuli in CIELUV ($L^*u^*v^*$) space. Experiments 1 and 2 used saturated stimuli and a saturated color wheel. Experiment 3 used a combination of saturated and desaturated color stimuli with a saturated color wheel. Clockwise from top left: (a) desaturated four-element display; (b) desaturated six-element display with desaturated color wheel: (c) saturated display with saturated color wheel, and (d) retrieval cue and post-trial feedback arc.

Method

Stimuli and Design

We report data and model fits for versions of the WC-CDM from three continuous-outcome hue identification experiments that varied the maximum set size, $m_{\max}$= 4 and $m_{\max}$= 6, and the color saturation of the stimuli. All experiments used a mouse-movement decision task, in which subjects responded by moving a mouse from a central home position to a point on the surrounding color wheel, as shown in Figure 4. Full details of the experiments can be found in Appendix B; we limit ourselves here to those features that

are relevant to the modeling of data and the evaluation of theory. All experiments used small-N designs in which several thousand experimental trials were collected from each of five subjects and all of the modeling was done at the individual subject level (Smith & Little, 2018). These designs provide the high level of measurement precision needed to test the marginal and joint distribution predictions of the models, as remarked previously.

Experiments 1 and 2 are mouse-movement versions of Experiments 1 and 2 in Smith et al. (2024/2026), whose JP parameters are plotted as Saa23 and Saa23a in Figure 2a. Those experiments used an eye-movement decision task, in which subjects responded by moving their gaze from a fixation point to a point on the surrounding color wheel. Subjects in our experiments here made a ballistic mouse movement from a home circle to a point on the color wheel. Experiments 1 and 2 used saturated stimuli and a saturated color wheel and varied the maximum set size ($m_{\max}$= 4 and $m_{\max}$= 6, respectively), for the reasons discussed earlier — namely, that smaller set sizes limit the effects of guessing, which is useful when attempting to identify the scaling law that determines the set size effect. We chose to replicate those experiments here because the retrieval precision in Smith et al.'s $m_{\max}$= 6 experiment was lower than in their other experiments and the set-size function followed a power law with an exponent of $\beta \approx 0.4$, which is less than sample-size. Smith et al. attributed the lower precision and smaller exponent to measurement noise in the eye-movement recordings. To ensure we had comparable data for the $m_{\max}$= 4 and $m_{\max}$= 6 experiments we replicated them using a mouse-movement task. All experiments used a backwardly-masked, 1,500 ms stimulus exposure followed by a 1,000 ms retention interval. At the end of the retention interval, a single location was probed for recall, as shown in Figure 4. Trial-by-trial accuracy feedback was given by means of an arc that showed the discrepancy between the true and reported hue of the probed item on the color wheel, as shown in Figure 4d.

Experiment 3 is a reanalysis of Experiment 3 in Smith et al. (2024/2026), whose JP ψ parameters are plotted as All24 in Figure 2. This experiment used $m_{\max}$= 6 and varied the color saturation of the stimuli. The aim of this manipulation was to distinguish features of the model that characterize the difficulty of the decision task, independently of those that determine the quality of the information retrieved from memory. The continuous-outcome decision task requires the hue of an item retrieved from memory to be matched to the hue of a corresponding point on the color wheel. Varying the saturation of the stimuli varies the difficulty of the decision task because it varies the amount of the primary hue in the stimulus, which is the attribute used to make the decision. The logic of this manipulation is the same as that of other perceptual decision tasks in which difficulty is manipulated by varying the proportion of a relevant stimulus attribute.

Experiment 4 of Smith et al. (2024/2026) extended the logic of their Experiment 3, by factorially varying the color saturation of the stimuli and the color wheel, with $m_{\max}$=

4. Subjects made decisions about saturated and desaturated stimuli and expressed them on saturated and desaturated color wheels. The results showed that the decision task was easier and RTs were shorter when the saturation of the stimuli matched that of the color wheel and harder when whey mismatched. The task was hardest and RTs were longest with desaturated stimuli and a saturated wheel. These results parallel those of those Tomić et al. (2024) who found little effect of saturation on retrieval accuracy when the saturation of the stimulus matched that of the color wheel. We did not reanalyze the results of Smith et al.'s Experiment 4 here because their conclusions about the set-size effect closely paralleled those for Experiment 3.

Fitting the Models to Data

We fitted the models to the individuals subjects' error and RT data using maximum likelihood estimation using the methods described in Smith et al. (2020, 2022). The predictions were generated using a combination of Matlab and C, again as described by Smith et al., but making use of a correction factor described by Smith et al. (2023) to stabilize them numerically for large values of drift rate and decision criteria. For each experiment we fitted the model by minimizing twice the log-likelihood ratio ($-2LL$) using a combination of a particle swarm optimizer and the Nelder-Mead simplex algorithm, implemented in Matlab as `particleswarm` and `fminsearch` respectively.

Table 1 Parameters of the Decision Models

Parameter	Symbol
Criterion circle radius	a
Mean drift rate norm	$\|\boldsymbol{\nu}\|$
Wrapped Cauchy memory dispersion	κ_μ
Drift rate radial variability	η
Power law exponent	β
Mean nondecision time	T_{er}
Nondecision time variability	s_t
AWSS model unattended item weight	u
AWSS model attention span	k_m

Note. $\sigma^2 = 1$ for all conditions

Smith et al. (2024/2026) considered highly parameterized models, which sought to capture the anisotropies associated with different parts of the stimulus space (Bae et al., 2015; Hardman et al., 2017; Kvam et al., 2023; Persaud & Hemmer, 2016; Smith et al. 2020) and to characterize the speed and accuracy of decisions for $m = 1$ trials, on which the probed item is known at the start of the retention interval rather than the end. VWM studies often omit the $m = 1$ condition to avoid the particular features of these decisions,

but we included them because of our interest in the scaling law that governs the set-size effect, which is a function of single-item performance. Here we take the opposite approach to the one in Smith et al. and present the most parsimonious models that can characterize the main features of the data. These models require fewer than half the number of parameters of those reported by Smith et al., but nevertheless capture the main features of the marginal and joint distributions of errors and RT and how they change with set size and color saturation. Table 1 lists the parameters of the WC-CDM and its AWSS special case. The numbers of parameters of each kind varied with the experimental design and are described in what follows.

The parameters fall into four general classes: parameters of the stimulus representation in memory; parameters of the AWSS model; parameters of the decision process, and nondecision time parameters. Each stimulus is represented by a distribution of drift rates with a mean drift rate (drift rate norm), $\|\boldsymbol{\nu}\|$, and radial and polar components of drift rate variability. The polar component is represented by a wrapped Cauchy distribution with dispersion κ_μ; the radial component is normally distributed with standard deviation, η. We compared power law and AWSS models of how κ_μ varies with set size. The general power law with exponent β is atheoretical but includes the sample-size model with $\beta = 0.5$ as a special case. The AWSS model is a theoretically motivated alternative to the power law which holds that large exponents are a reflection of unequal allocation to evidence samples to items. The k-item focus of attention model is characterized by two parameters: the unattended item weight, u, and the attention span, k_m. It includes the single-item focus of attention model of Smith et al. (2018) with $k_m = 1$ as a special case.

The only unique parameter of the decision process is the radius of the criterion circle, a. As the set sizes in our experiments were randomized across trials within blocks we made the selective influence assumption that the same decision criterion is used for all stimuli. Fennell and Ratcliff (2023), in their application of the SCDM to VWM, allowed the decision criterion to differ as a function of set size, arguing that the retention interval provides enough time to allow different criteria to be set for displays of different sizes. Although this argument is defensible it involves abandoning one of the core selective influence assumptions of evidence accumulation models. We found we could successfully fit the CDM while retaining selective influence.

We modeled nondecision time as an independent, uniformly-distributed random variable with mean T_{er} and range s_t. Although other models of nondecision times have been proposed by different authors (see Smith & Ratcliff, 2025, for a review), the additive, independent decomposition of decision and nondecision times has the advantage that it is simple, involves the fewest auxiliary assumptions, and has been shown to work well in practice. Many studies have found that nondecision times remain relatively invariant across stimulus conditions, but in VWM, nondecision times often vary with set size. Sewell et al.

(2016) and Smith et al. (2016) found that T_{er} progressively increased with set size in fits of the two-choice diffusion model; Smith et al. (2024/2026) found the same pattern in fits of the JP-CDM. Sewell et al. linked the increase in nondecision time to the idea an item in VWM must be brought into the focus of attention before it can be retrieved and this takes longer when there are more items in memory (Oberauer, 2002; Shepherdson, 2020; Shepherdson et al., 2017). We also allowed T_{er} to vary with set size, but constrained it to be constant across manipulations of saturation in Experiment 3. We assumed that s_t was constant across both set size and saturation in all experiments.

Results

Experiments 1 and 2

Table 2 Model Parameters, Experiments 1 and 2

Model	Parameters
$(1\nu, 4\kappa)$	1 drift rate, 4κ, free precision
$(1\nu, 1\kappa, \beta)$	1 drift rates, 1κ, power law precision
$(1\nu, 1\kappa, 0.5)$	1 drift rates, 1κ, sample-size precision
$(1\nu, 1\kappa, u, k_m)$	1 drift rates, 1κ, AWSS precision
$(1\nu, 1\kappa, 0, k_m)$	1 drift rates, 1κ, encoding failure

One day experiments will grow here....

Experiment 3

In Experiment 3, the set-size manipulation was crossed with a manipulation of stimulus saturation, using color patches like those in Experiments 1 and 2. In Experiment 3, the color patches were either saturated, within the limit of the monitor gamut, or desaturated. Desaturated colors appear duller, or more washed-out, than their saturated counterparts. The desatuated colors were constructed by mixing the saturated colors with grays of equal luminance, which shifted them towards the center of isoluminant plane. Set size and color saturation trials were randomly mixed within an experimental block. There were no mixed saturation trials: Stimuli on any trial were either all saturated or all desaturated. We conducted a replication of this experiment (Jg26 in Figure 2a), in which the saturation manipulation was blocked rather than randomized, with no essential difference in the results. The purpose of the saturation manipulation was to vary the difficulty of the decision task by varying the amount of the primary hue in the stimulus.

Table 3 summarizes the main models we considered here. Smith et al. (2024/2026) considered several more complex models, but the main experimental effects of interest can be characterized by the models in the table. All of the models but one had two mean drift

Table 3 Model Parameters, Experiment 3

Model	Parameters
$(2\nu, 8\kappa)$	2 drift rates, 8κ, free precision
$(2\nu, 2\kappa, \beta)$	2 drift rates, 2κ, power law precision
$(1\nu, 2\kappa, \beta)$	1 drift rate, 2κ, power law precision
$(2\nu, 2\kappa, 0.5)$	2 drift rates, 2κ, sample-size precision
$(2\nu, 2\kappa, u, k_m)$	2 drift rates, 2κ, AWSS precision
$(2\nu, 2\kappa, 0, k_m)$	2 drift rates, 2κ, encoding failure

rates, $\|\boldsymbol{\nu}\|$, one radial drift rate standard deviation, η, four mean nondecision times, T_{er}, and one nondecision time range, s_t. The exception was $(1\nu, 2\kappa, \beta)$, the power law model with a single mean drift rate, which we included as a benchmark to show the need for different drift rates for high and low saturation conditions. Smith et al. considered models in which drift rates were free to vary both with saturation and set size and found there was a tendency for drift rates to decrease with set size and for the difference in drift rates for high and low saturation stimuli to decrease with set size, but these models were not preferred to simpler models by the model selection statistics.

The main differences in the models in the table is in how they parameterized the set-size effect. The models $(2\nu, 2\kappa, 0.5)$, $(2\nu, 2\kappa, \beta)$,and $(2\nu, 8\kappa)$, are the sample-size, power law, and free-κ models, respectively. The first two models allowed κ_1 for $m = 1$ trials to vary with saturation and constrained κ for larger set sizes to follow a sample-size or power-law. The free κ model allowed precision to vary freely as a function of both set size and saturation. Model $(2\nu, 2\kappa, u, k_m)$ is the AWSS model, with unattended attention weight u and attention span k_m. Model $(2\nu, 2\kappa, 0, k_m)$ is the encoding failure model, in which items outside of the span of attention have zero weight.

Figure 5a shows the rank ordering of the models in Table 3, averaged across subjects. Figure 5b through 5d compare the predicted precision (1/CSD) of the sample-size, power law, and AWSS to the predicted precision of the free κ model. As for Experiments 1 and 2, the sample-size model performed more poorly than the power law model, which performed similarly to the free κ model. The power law model with only a single mean drift rate for saturated and desaturated stimuli performed very poorly. Figures 5b and 5c show that a power law with an average exponent of $\beta = 0.66$ predicted a set-size effect similar to that of the free-κ model, but the sample-size model underestimated it.

The best model, by a small margin, was the AWSS model $(2\nu, 2\kappa, u, k_m)$. The average parameters of the AWSS model are shown in Table 4. The average attention weight and attention span of the AWSS model were $u = .197$, $k_m = 4.416$. These estimates imply that the differences between the sample-size and AWSS models really only become apparent on $m = 6$ trials. The performance of the AWSS model was essentially indistinguishable from

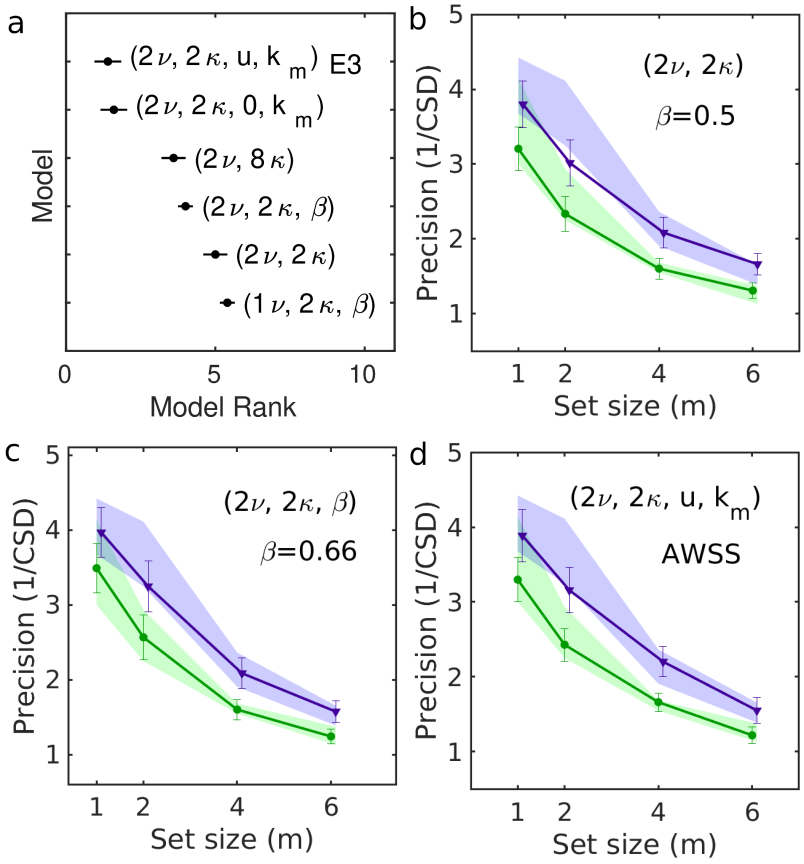


Figure 5: Model fits for Experiment 3. (a) Rank-ordered AIC fits, averaged across subjects. The error bars are standard errors of the averaged ranks. (b) through (d). Precision estimates (1/CSD) for the fitted models. The error bands are precision estimates for the free-κ model; the solid lines and symbols are estimates for the constrained models. The green bands and symbols are estimates for the low-saturation condition; the purple bands and symbols are estimates for the high-saturation condition. (b) $(2\nu, 2\kappa, 0.5)$, sample-size model; (c) $(2\nu, 2\kappa, \beta)$, power law model; (d) $(2\nu, 2\kappa, u, k_m)$, AWSS model. The error bars are 1 SEM.

that of the encoding failure model, for which $u = 0$. For this model, the estimated span of attention was slightly larger, $k_m = 5.104$. Compared to the AWSS model,the encoding failure model reduces the weighting of unattended items, but estimates fewer of them.

Table 4 Average parameters of the AWSS model, Experiment 3

$\|\boldsymbol{\nu}\|^-$	$\|\boldsymbol{\nu}\|^+$	κ_1^-	κ_1^+	η	u	k_m
3.944	4.240	0.054	0.027	0.850	0.197	4.416
a	$T_{\text{er},1}$	$T_{\text{er},2}$	$T_{\text{er},3}$	$T_{\text{er},4}$	s_t	
4.224	0.236	0.448	0.495	0.542	0.334	

Note. Superscripts "−" and "+" denote low and high saturation conditions, respectively. Subscripts index set sizes: $m = 1$, 2, 4, and 6.

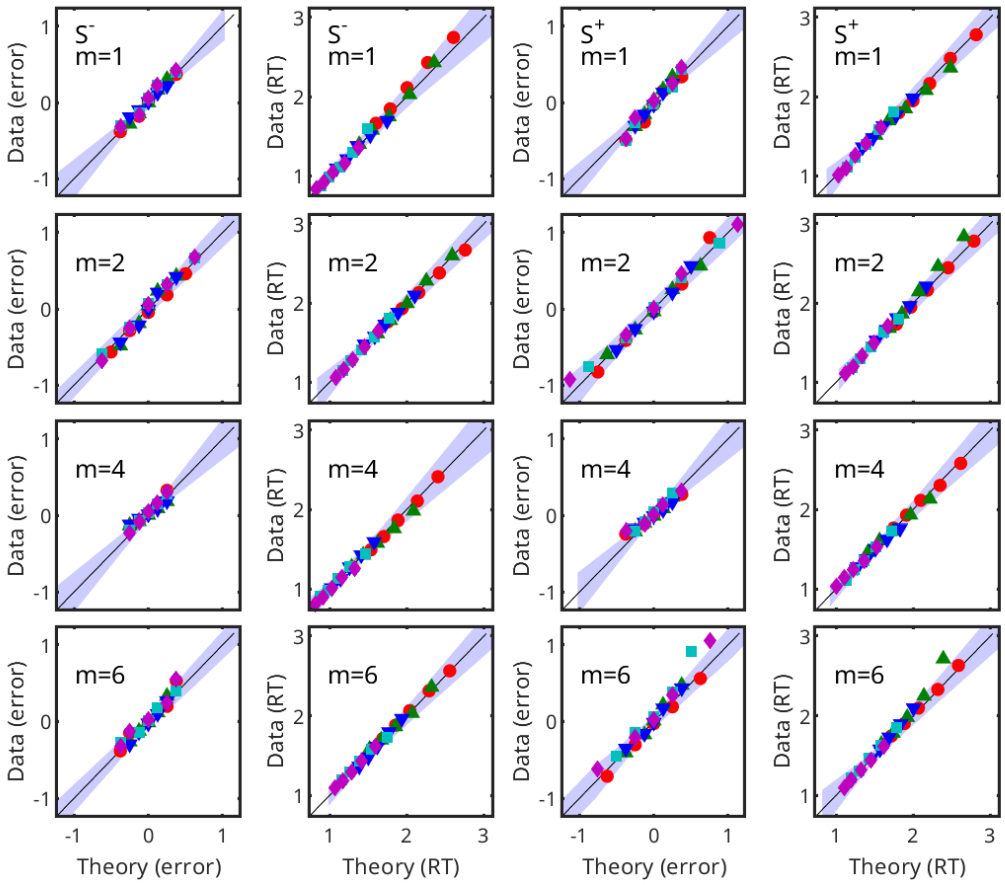


Figure 6: Q–Q plots of empirical distribution quantiles (Data) against fitted quantiles (Theory) for errors (left-hand panels) and RTs (right-hand panels) for all five subjects in Experiment 3 for the AWS model, $(2\nu, 2\kappa, u, k_m)$. The pair of columns on the left are errors and RTs for low saturation stimuli (S^-); the pair of columns on the right are errors and RTs for high saturation stimuli (S^+). The data points for individual subject are plotted in different colors with different plotting symbols. The shaded regions are $\pm 2SD$ bootstrap estimates from the empirical data.

Appendix A
The Wrapped Cauchy CDM

Cauchy and Wrapped Cauchy Distributions

The Cauchy distribution of a random variable Y with center μ and dispersion κ has probability density function

$$f(y) = \frac{\kappa}{\pi\left[(y-\mu)^2 + \kappa^2\right]}, \tag{A1}$$

(Applebaum, 2004, p. 34; Mardia & Jupp, 2000, p. 51). The wrapped Cauchy distribution of a circular random variable θ, where $-\pi \leq \theta < \pi$, can either be expressed as an infinite series

$$f(\theta) = \sum_{n=-\infty}^{n=\infty} \frac{\kappa}{\pi\left[\kappa^2 + (\theta - \mu + 2\pi n)^2\right]}, \tag{A2}$$

or in the simpler form of Equation 9 in the text (Jones & Pewsey, 2005; Mardia & Jupp, 2000, p. 51). For our purposes, a key property of the Cauchy and wrapped Cauchy distributions are that they infinitely divisible. This means that a sum of m random variables, Y, with center μ and dispersion parameter κ is distributed as a Cauchy random variable with center $m\mu$ and dispersion $m\kappa$. This property is most simply verified by considering the characteristic function (Fourier transform) of Y, which is defined as the expected value $E[e^{i\lambda y}]$ where λ is the complex frequency variable of the Fourier transform and $i = \sqrt{-1}$. It may be shown using complex variable theory that the characteristic function of the Cauchy distribution is

$$E[e^{i\lambda y}] = e^{i\mu\lambda - \kappa|\lambda|}, \tag{A3}$$

(Applebaum, 2004, p. 34). Like the Laplace transform, the Fourier transform converts convolution into multiplication. This means that the characteristic function of a sum $\sum_i Y_i$ of Cauchy random variables is simply the product of their characteristic functions. For m such identically distributed random variables,

$$E\left[\exp(i\lambda \sum_{i=1}^{m} y_i)\right] = \left(e^{i\mu\lambda - \kappa|\lambda|}\right)^m = e^{im\mu\lambda - m\kappa|\lambda|}. \tag{A4}$$

The term on the right-hand side can be recognized as the characteristic function of a Cauchy random variable with center $m\mu$ and dispersion $m\kappa$. When the associated probability distribution is wrapped around a circle it yields a version of Equation 9 in the text,

$$f(\theta; \kappa, \varphi) = \frac{\sinh m\kappa}{2\pi[\cosh m\kappa - \cos(\theta - m\varphi)]}. \tag{A5}$$

For the circular distribution we have changed the variable to θ and the center to φ. In the text the dispersion is denoted as κ_μ as a reminder that it is the dispersion of the drift rate in the CDM. Mardia and Jupp (2005, p. 51) and Jones and Pewsey (2005) give alternative forms of Equation A5 that are parameterized in terms of $\rho = e^{-\kappa}$ and $\tanh(\kappa/2)$, respectively.

Subordinated Brownian Motion

The key to the wrapped Cauchy distribution in the CDM is the doubly-stochastic nature of the VWM consolidation process that determines the polar angle of the drift rate distribution. We model this as a subordinated Brownian motion process. This process, Y_t^2, runs for a random time, determined by the first-passage properties of a second Brownian motion process, Y_t^1, which is termed the Lévy subordinator. The doubly-stochastic nature of the subordinated process leads straightforwardly to a Cauchy distribution (Mörters & Peres, 2010, p. 53). The distribution of the polar angle of the VWM trace at the consolidation time, T, is obtained by integrating the distribution of the memory process, Y_t^2, over the distribution of random times of the subordinator, Y_t^1,

$$\begin{aligned} f_T(y) &= \int_0^\infty g_T(t) g_{Y_2}(t)\, dt \\ &= \int_0^\infty \frac{\gamma}{\sqrt{2\pi t^3}} \exp\left(-\frac{\gamma^2}{2t}\right) \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left[-\frac{(y-\varphi)^2}{2\sigma^2 t}\right] dt. \end{aligned} \tag{A6}$$

In Equation A6, we assume the diffusion coefficient of the Y_t^1 process, given by Equation 12, is unity, but allow the diffusion coefficient of the Y_t^2 process, denoted σ^2, to vary. The variable φ is the center of the Y_t^2 process and depends on the encoded stimulus representation; it is not a drift rate and has no time-dependent term. Any diffusion that occurs during the consolidation time is symmetrical and drift-free. We evaluate the integral in Equation A6 by making the substitution

$$z = \frac{\gamma^2\sigma^2 + (y-\varphi)^2}{2\sigma^2 t}$$

with

$$dt = -\frac{\gamma^2\sigma^2 + (y-\varphi)^2}{2\sigma^2 z^2} dz$$

to obtain

$$f_T(y) = -\frac{\gamma\sigma}{\pi[\gamma^2\sigma^2 + (y-\varphi)^2]} \int_0^\infty e^{-z}\, dz = \frac{\gamma\sigma}{\pi[\gamma^2\sigma^2 + (y-\varphi)^2]}. \tag{A7}$$

The right-hand side of Equation A7 can be recognized as the probability density function of a Cauchy distribution with dispersion parameter $\gamma\sigma$. The boundary γ in Figure 3, which determines how far the Y_t^1 process has to travel before consolidation occurs, determines the dispersion of the VWM memory representation, along with σ^2, the diffusion coefficient of Y_t^2. Rather than assuming that γ is a variable "criterion," as is usually the case in cognitive applications of Equation 12 (e.g., Smith & Ratcliff, 2025, p. 297), we assume it is a hard-wired property of the memory system that characterizes a threshold level of activation needed to consolidate the item representation in VWM. In the WC-CDM, the polar component of the drift rate distribution is wrapped around a circle, as shown in

Figure 3. We denote the dispersion of the wrapped Cauchy as κ_μ in Equation 9. The circular standard deviation of the drift rate distribution is $\kappa_\mu = \sqrt{2\gamma}\sigma$, by Equation 11. In fits of the model, γ and σ are not independently identifiable; the relevant parameter is the dispersion parameter κ_μ and how it varies with set size.

Sample-Size and Attention Weighted Sample-Size CDM

The sample-size model views VWM as a resource comprising a set of noisy evidence samples that are recruited during the exposure period and are available to represent stored items. The more samples that are allocated to an item, the more precise its representation will be. In the normal signal detection model, this leads to the inverse square-root relation in Equation 13. In the WC-CDM, we conceive of the samples as a set of n independent copies of the Y_t^2 process, each with diffusion coefficient σ^2, where the value of n varies with exposure duration. The memory representation of an item is based on an average of the copies that represent it. The average of n Brownian motion processes with diffusion coefficient σ^2 is itself a Brownian motion with diffusion coefficient $\sigma_n^2 = \sigma^2/n$. When there is only one item in the display, all n samples (Y_t^2 processes) will be available to represent it and the dispersion of the wrapped Cauchy distribution will be $\kappa_1 = \gamma\sigma/\sqrt{n}$ (where the subscript now denotes the number of items in memory). When there are m items in the display and the samples are equally divided among them, such that each item receives n/m samples, the dispersion will be $k_m = \gamma\sigma/\sqrt{(n/m)}$, yielding $\kappa_m = \sqrt{m}\kappa_1$, which is the sample-size model. Smith (2024/2026) assumed this relationship in their fits of the JP-CDM, but for the WC-CDM it is exact. The attention-weighted form of the model, described in the text, assumes that the allocation of samples/processes to items is unequal rather than unequal, as characterized by Equation 14.